
Multi-Objective Reinforcement Learning for Carbon-Aware Global Logistics

CSCI-6364 S26 - Machine Learning

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Abstract

The goal of this project is to create a carbon-centric logistics routing engine that utilizes Multi-Objective Reinforcement Learning to optimize for the reduction of delivery time as well as carbon emissions in the context of freight transport. Existing GPS-based logistics systems only focus on the optimization of speed without considering the high variability in fuel consumption due to factors like traffic congestion, road grade, and vehicle load. In the proposed system, the road network is represented as a dynamic graph, where the weights of the edges represent the actual Carbon Intensity, which is the combination of factors including vehicle emission factors, road grade, traffic speed, and time of day. An RL agent is trained on a representative DMV subgraph to navigate the Pareto frontier between delivery speed and CO₂ emissions, with Weighted A* on carbon weights as a validated fallback. We hypothesize measurable CO₂ reduction relative to the fastest route baseline, which we will quantify in our evaluation.

Background

The global logistics industry accounts for approximately 8% of worldwide greenhouse gas emissions, with road freight alone producing more than 2 billion metric tons of CO₂ annually (IEA, 2023). Despite this, popular routing systems, like Google Maps and Waze, are primarily focused on minimizing travel time, ignoring the substantial emissions caused by traffic, terrain, and vehicle type.

The regulatory landscape around emissions is changing quickly. Initiatives such as the EU's Carbon Border Adjustment Mechanism (2023) and proposed U.S. EPA Scope 3 emissions reporting indicate that companies will soon be financially penalized for logistics-related carbon output. Additionally, advances in research have produced powerful tools, like Reinforcement Learning, and open geospatial datasets, which make it possible to approach this kind of problem for the first time.

Theory of Change

Motivation

The current state of the software used in logistics takes CO₂ emissions as an afterthought. For example, a driver routed through DC during rush hour may idle for 40 minutes, producing 3–5 times the CO₂ of an alternate route. This, again, was not a problem that was taken into consideration, but is now made obvious by the Scope 3 emissions report.

The inspiration for this project arises from the realization that, for instance, the "fastest route" and the "greenest route" may only differ by 5-10 minutes, yet they may differ significantly in terms of their fuel consumption. A model that can achieve this, and explain the difference to a logistics manager, is immediately useful.

If successful, the project would teach us how to integrate mixed real-world data sources into single ML-ready graph structures. It would teach us how to apply RL to routing under competing objectives and how to define the reward functions that navigate a Pareto frontier rather than optimizing a single metric. It

will also expose the limitations of DQN scalability on large graphs and how noisy real-world traffic and emissions data is in practice.

Survey of Existing Work

The field of Green Vehicle Routing has a substantial body of prior work, which our project builds upon:

Erdoğan & Miller-Hooks (2012) introduced the GVRP as a formal extension of the classic Vehicle Routing Problem, adding fuel station constraints and emission objectives. This paper establishes the foundational mathematical framework our edge weight design is based on. A comprehensive update is provided by Garside, Ahmad & Muhtazaruddin (2024), whose review of 458 GVRP papers from 2016-2023 presents that multi-objective extensions now dominate the field. This directly validates our choice of Pareto-based multi-objective routing for balanced delivery time and CO₂.

Zou, Wu, Yin et al. (2024) present an improved Transformer model that combines multi-head attention and attention-to-attention mechanisms to solve the low-carbon multi-depot vehicle routing problem. This demonstrated that end-to-end deep RL with an encoder-decoder framework outperformed traditional methods including genetic algorithms and Google OR-Tools on carbon constrained routing.

Yue et al. (2024) addressed the time dependent green VRP with time windows. They proposed a two-stage DQN-based multi-objective algorithm that optimizes both travel time and carbon emissions. Additionally, it also optimizes customer satisfaction under traffic and time-varying conditions. Their use of

Pareto-front strategies to balance competing objectives under congestion is directly analogous to our reward scalarization approach, and their results confirm that DQN is effective for this class of problem.

Wang, Qiu, He, Zhou & Strbac (2023) proposed a multi-agent RL method with a hierarchical hybrid policy to optimize EV eco-routing and carbon emission reduction. This was evaluated on real-world central London transport networks. This study confirms that RL-based carbon routing scales to real road networks and achieves measurable emission reductions without having prior knowledge of the system.

The Virginia Tech Comprehensive Power-Based Fuel Consumption Model (VT-CPFM) by Rakha et al. (2011) provides the physics-based foundation informing our Carbon Weight design, modeling fuel consumption as a function of vehicle speed, acceleration, and road grade.

Formulation and Scope

Problem Formulation

We model the logistics network as a directed weighted graph $G = (V, E)$, where nodes V represent warehouses, hubs, and delivery stops, and edges E represent OSM road segments. Each edge carries a dynamic Carbon Weight adapted from vehicle dynamics emission modeling [Wang et al., 2023; Rakha et al., 2011]:

$$CW(e) = \text{distance}(e) \times EF(k) \times \text{grade_factor}(e) \times \text{congestion_factor}(e, t)$$

$EF(k)$ is the EPA emission factor for vehicle class k (g CO₂/mile). grade_factor is computed as rise/run from SRTM elevation data attached to each node. congestion_factor applies a two-period peak/off-peak speed adjustment (0.6× during 7–9am and 4–7pm) calibrated from FHWA TMS data, with OSMnx road-classification imputation as fallback for uncovered segments.

Model Architecture

The graph is constructed using OSMnx for the DMV road network, enriched with SRTM elevation. EPA emission factors are mapped per edge for two vehicle classes: Class 8 Diesel Truck and Electric Van. The DQN agent represents state as a 64-dim node embedding concatenated with a time-of-day bin, with actions selecting the next neighbor node and reward as scalarized negative normalized travel time and CO₂. Training is staged: Stage 1 on a ~1,000-node downtown DC subgraph; Stage 2 scales to full DMV only if Stage 1 converges, otherwise Weighted A* on carbon weights serves as the full-network method.

Baselines are Dijkstra (fastest path) and greedy minimum-emission. Scope is limited to the DMV metro area, batch-processed, with no live API integration.

Milestones

Weeks 1–2 focus on data collection and graph construction: pulling the DMV road network from OpenStreetMap, attaching SRTM elevation to each edge, and imputing speed profiles via OSMnx. The deliverable is a carbon weight heatmap overlaid on the DMV road network.

Weeks 3–4 are spent building and validating the emission model. We apply the carbon weight formula across edge types and check outputs against EPA emission factor tables, with a side-by-side comparison of diesel truck vs. electric van profiles.

Weeks 5–6 are the first RL iteration. We stand up the DQN agent on a ~1,000-node downtown DC subgraph, define the state and action space, tune the scalarized reward, and debug training. The deliverable is a working agent on the subgraph with an initial route comparison against Dijkstra.

Weeks 7–8 are evaluation. We run the trained agent across 50 sampled origin-destination pairs and quantify CO₂ savings and time penalty relative to the fastest-route baseline. If Stage 1 converged cleanly, we attempt to extend to the full DMV graph. The final report documents both outcomes regardless.

Uncertainties and Back-Up Plans

Risk 1: RL Training Instability on Large Graphs: RL agents are notoriously difficult to train on large, sparse graphs. The state space for the full DMV road network could be too large for a naive DQN to converge in the project timeline.

Back-up Plan: Reduce scope to a representative subgraph of around 1,000 nodes covering a single county. Alternatively, replace the RL agent with a simpler Weighted A* pathfinder using our carbon weights, which still demonstrates the core contribution of carbon-aware routing without the RL complexity.

Risk 2: Traffic Speed Data Gaps: Hourly speed data might not be available for all segments in Virginia and Maryland through the DC Open Data website.

Back-up Plan: Use the Microsoft GeoLife dataset (GPS trajectories in major cities)/ OSMnx road-classification based speed imputations or synthesize congestion factors from OpenStreetMap speed limits with a time-of-day degradation curve derived from published traffic engineering studies.

Risk 3: Carbon Weight Calibration: The `grade_factor` and `congestion_factor` terms in our carbon weight formulation require calibration against real emission measurements. Without validation data, the model may produce carbon estimates that are directionally correct but numerically inaccurate.

Back-up Plan: Normalize all carbon weights relative to a baseline such as flat roads, so results are reported as relative CO₂.

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